DEEPFAKE X — FORENSIC MEDIA INTELLIGENCE REPORT

STRUCTURED FORENSIC EVIDENCE EXAMINATION & METADATA AUDIT

Investigation ID: TEST-MANIFEST-001	**Date/Time:** 2026-11-12 00:00:00 UTC
Case Title: Manifest Integrity Verification	**Investigator:** Forensic Officer
Overall Assessment: Potential manipulation detected	**Custody Standard:** Structured Forensic Evidence Handling

1. EXECUTIVE EVIDENTIARY ASSESSMENT

Forensic Category: Potential manipulation detected
Evidentiary Strength: Single modality alert

Technical Rationale: Single image flagged.

2. EVIDENCE INVENTORY & CHAIN OF CUSTODY (SHA-256)

Filename	Modality	Size	SHA-256 Digest
test.png	IMAGE	100 KB	abcdef1234567890

3. DETECTOR INFERENCE FINDINGS (UNCALIBRATED SOFTMAX SCORES)

Image Analysis (Swin Transformer Vision Backbone):
• **test.png**: AI-Generated Image | Synthetic: 95.0% | Authentic: 5.0% | Device: cuda:0

4. METADATA & CONTAINER COMPARISON MATRIX

No cross-modal pairs evaluated.

5. GROUNDED FORENSIC EXPLAINABILITY & OPERATIONAL BOUNDS

No detailed explainability outputs available for this case.

6. ADVANCED EVIDENCE COMPARISON & PROVENANCE LINEAGE

A. Cryptographic Evidence Lineage (Strict Provenance):
• No verified parent-child evidence lineage detected (all media items ingested independently).

7. INVESTIGATION AUDIT TIMELINE ANALYTICS

Ledger Sequence Integrity: VALID (Cryptographically Intact) | **Total Events Logged:** 0 | **Investigation Span:** 0.0s
Cryptographic Integrity Notice: Hash-chain validation confirms that no audit entry was inserted, removed, or altered out of sequence. It does not verify host system clock accuracy.

8. FORENSIC METHODOLOGY & EMPIRICAL DETECTOR LIMITATIONS

Forensic Evidentiary Disclaimer: This report is generated strictly for investigative screening. Model scores reflect raw uncalibrated softmax activations of deep neural feature extractors and do not represent absolute mathematical truth.
1. **Image Detection (Swin Transformer):** Susceptible to false negatives on novel generative diffusion and flow architectures.
2. **Video Screening:** Performs uniform frame-level screening; does not constitute face-swap or lip-sync biometric proof.
3. **Audio Detection (Wav2Vec2):** Known empirical vulnerability to modern neural flow-matching speech synthesis (e.g., ElevenLabs). Evaluated accuracy is vocoder-dependent; false negatives occur on state-of-the-art voice clones.
4. **Cross-Modal Comparison:** Pairwise container comparison does not prove common origin or intentional manipulation without independent contextual corroborate.